

1) A pair of 3×3 convolution kernels of sobel operator (G_x and G_y) as shown in the below figures.

Solution → Given image $\begin{pmatrix} 12 & 10 & 13 \\ 14 & 12 & 13 \\ 15 & 14 & 11 \end{pmatrix}$

Sobel kernels

$$G_x = \begin{pmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix}$$

$$G_y = \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$G_y = (-1 \times 12) + (-1 \times 10) + (-1 \times 13) + (15 \times 1) + (14 \times 1) + (11 \times 1)$$

$$G_y = G_x = (-1 \times 12) + (0 \times 10) + (1 \times 13) + (-1 \times 14) + (12 \times 0) + (13 \times 1) + (-1 \times 15) + (14 \times 0) + (1 \times 11)$$

$$= 9$$

Gradient magnitude

$$G = \sqrt{(G_x)^2 + (G_y)^2}$$

$$= \sqrt{(5)^2 + (9)}$$

$$= 10.3$$

2) Explain the region-based Approach for image segmentation with proper example.

Solution

The region based approach for image segmentation divides an image into meaningful regions based on predefined criteria such as intensity, color, texture, or connectivity. This approach groups similar pixels together and separates dissimilar ones to extract objects or areas of interest from an image.

Key techniques in
Region Based Segmentation

1) Region Growing

This technique starts with a seed pixels and expands the region by adding neighbouring pixels that have similar properties
(Example - intensity or color)

Example

image →

5	6	8	100	102	101
7	8	9	98	101	99
6	7	9	97	100	98
5	6	8	96	99	97

Step 1

select an initial seed pixel based on a criterion (eg - intensity)

→ select seed pixel (example top left pixel with intensity 5)

Step 2 → compare the seed pixel with its neighbours.
→ grow the region by adding pixels with similar intensity (values near 5-9)

Step 3 If the neighbouring pixel meets the similar criterion add it to the region.
- Ignore higher-intensity pixel (96-102) since they belong to another region.

Step 4 Repeat the process until no more pixels meet the condition.

→ The output is two segmented regions:

(a) dark background

(b) bright object

ii) Region Splitting and Merging

This technique recursively splits the image into smaller regions until each region is homogeneous. If neighbouring regions are similar, they are merged.

Example If an image has different intensity areas

Step 1 Start with the entire image as one region

3) Compare and contrast global and local registration techniques in detail.

Solution → Image registration is the process of aligning two or more images of the same scene taken at different times, from different viewpoints, or using different sensors. The two main types of registration techniques are global registration and local registration.

Global Registration Characteristics

- i) Uses a single transformation function for the whole image.
- ii) Suitable for images with minor distortion.
- iii) Computationally efficient.

Local Registration Characteristics

- i) Uses multiple localized transformations instead of a single global one.
- ii) Handles non-uniform distortions like warping and local misalignment.
- iii) Computationally complex.

④ Explain the dilation and Erosion techniques to perform the morphological operations with proper example.

Solution →

Dilation → Dilation expands the boundaries of objects in an image. It adds pixels to the objects edges, making it thicker.

Working process

- i) A structuring element is placed over the image.
- ii) If at least one pixel under the structuring element is 1, the corresponding output pixel is set to 1.
- iii) The operation is repeated over the entire image.

Example input binary image (3x3)

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Structuring element (3x3)

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

After dilation →

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

5 What is sharpening of an image and how it is done.

Solution → Image sharpening is a technique used in image processing to enhance edges and fine details in an image. It helps improve the clarity and visibility of important structures by increasing the contrast along edges.

How is image sharpening done?

Sharpening is typically done using spatial filter, median filter, minimum and maximum filter.

6 Explain four primary steps involved in the image registration process.

→ Image registration is the process of aligning two or more images of the same scene taken at different times, from different times, from different viewpoints, or using different sensors. The goal is to find a geometric transformation that maps one image (moving/floating image) onto another (fixed/reference image).

1) Feature detection

Identify key features (points, edges, corners, or regions) in the images that can be used for alignment.

Erosion → Erosion shrinks the boundaries of objects by removing pixels from the edges.

Working process

- i) The structuring element is placed over the image.
- ii) If all pixels under the structuring element are 1 the output pixels remain 1 otherwise it set 0.
- iii) The operation is repeated over entire image.

example input binary image (5×5)

0	1	1	1	0
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
0	1	1	1	0

Structuring element (3×3)

1	1	1
1	1	1
1	1	1

output image

0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

Common feature detectors

a) Corner Detectors → Harris Corner, FAST

(b) Edge detectors: Canny, Sobel

(c) Blob Detectors: SIFT

Example → In satellite imagery, feature detection can identify roads, buildings, or landmarks for alignment.

ii) Feature matching:

Establish correspondence between detected features in the reference and moving images.

7) What is the significance of the structuring element in morphological operations.

Solution → A structuring element is a small matrix (kernel) used in morphological operations to probe and modify the shape of objects in an image. It determines how pixels in the input image are processed and affects the results of operations like dilation, erosion, opening and closing.

Importance →

i) defines the shape of morphological changes

• The structuring element controls how an operation is applied to objects in an image.

Different shapes of structuring elements lead to different transformations of the image.

techniques

8 Explain hit and miss transformation, with proper example.

Solution → The hit or miss transformation is a morphological operation used to detect specific patterns or shapes in a binary image.

For a given binary image I , the Hit or Miss transformation is defined as

$$I \oslash S = (I \ominus S_1) \cap (\text{Complement}(I) \ominus S_2)$$

where

S_1 is the foreground structuring element

S_2 is the background structuring element

$\ominus$ → erosion

Example Consider input binary image

$$I = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Add a structuring element

$$S_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$S_2 = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

9) Describe three common techniques used for image segmentation. Explain how each technique works and provide a real life application for each.

Solution

Image Segmentation types

i) Threshold Based Segmentation

(a) divides image into background and foreground based on pixel intensity

(b) A threshold value T is chosen, and pixels are classified as —

$$I(x, y) = \begin{cases} 1, & \text{if } f(x, y) > T \text{ (object)} \\ 0, & \text{if } f(x, y) \leq T \text{ (background)} \end{cases}$$

Application

Document binarization

ii) Edge-Based Segmentation

(a) Detects edges where pixel intensity changes significantly

(b) Application

Autonomous Vehicle.

iii) Region Based Segmentation

(a) Groups pixels based on similarity in color, texture, or intensity.

- ⑥ Two popular approaches —
- region growing
 - Watershed Algorithm

Application → Medical Imaging

10> Explain transform and Similarity measures in details for registration with popular example.

Solution →

Transformation models in image registration →

A transformation model defines the mathematical function that maps points from the moving image to the reference image. Different transformation types vary in complexity and application.

Types of Transformation

- ① Rigid Transformation
- ② Affine Transformation
- ③ projective Transformation
- ④ NonRigid Transformation

Types of Similarities measures

- (a) mean squared error.
- (b) Cross Correlation
- (c) mutual information
- (d) Structural similarity index model

11 Explain how monomodel and multimodel image registration techniques perform image restoration. Show with proper example.

Solution →

i) Mono model Image Registration

Mono model registration is used when images come from the same sensor and have ~~some~~ similar intensity distributions but may have slight variation due to motion blur, lighting changes or noise.

ii) Multimodel image Registration

Multimodel image Registration is used when images come from different sensors or modalities. It helps combine information from different sources to restore missing details.

- 12) A pair of 3×3 matrix (convolution kernels) of sobel operator (G_x and G_y) given. Perform edge detection technique.

Image $\rightarrow$

3	1	4	2
3	2	5	1
2	3	4	5
3	4	5	6

Solution

$$G_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$

$$G_y = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix}$$

input image (1) =

3	1	4	2
3	2	5	1
2	3	4	5
3	4	5	6

window (2,2)

$$\begin{pmatrix} 3 & 1 & 4 \\ 3 & 2 & 5 \\ 2 & 3 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 4 & 2 \\ 2 & 5 & 1 \\ 3 & 4 & 5 \end{pmatrix}$$

window (2,3)

window (3,2)

$$\begin{pmatrix} 3 & 2 & 5 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 5 & 1 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}$$

window (3,3)

For (2,2) window

$$G_x = (-1) \times 3 + 0 \times 1 + 1 \times 4 + (-2) \times 3 + 2 \times 0 + 5 \times 2 + (-1) \times 2 + 0 \times 3 + 4 \times 1$$

$$G_y = 7$$

$$G_y = -3$$

Edge detected Output —

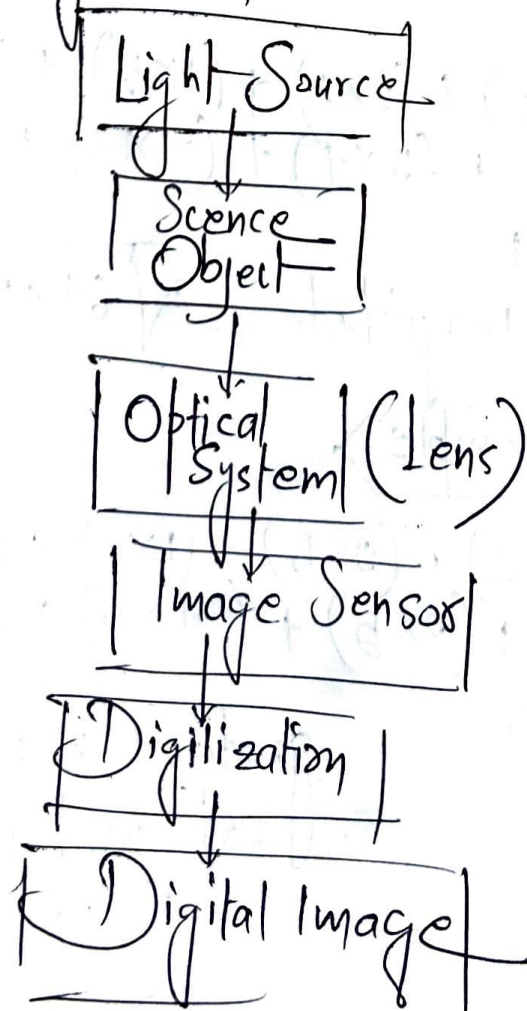
$$\begin{pmatrix} 7.6 & 8.1 \\ 8.9 & 8.6 \end{pmatrix}$$

(14) Explain Image formation model with diagram.

Solution → Components of image formation

- (i) Illumination (Light Source)
- (ii) Scene Object (Reflectance properties)
- (iii) Optical System (Camera)
- (iv) Sensor (Image detector)
- (v) Digitization

Diagram →



gradient magnitude

$$G = \sqrt{(7)^2 + (-3)^2} = 7.6$$

For (2,3) window

$$G_x = -1 \times 1 + 0 \times 4 + 1 \times 2 + -2 \times 2 + 0 \times 5 + 2 \times 1 \\ + (-1) \times 3 + 0 \times 4 + 1 \times 5$$

$$= 1$$

$$G_y = -5$$

gradient magnitude

$$= \sqrt{(1)^2 + (5)^2} = \sqrt{26} \\ = 5.1$$

For (3,4) Example

$$G_x = (-3) \times 2 + 0 \times 2 + 1 \times 5 + -2 \times 2 + 0 \times 3 + 2 \times 4 \\ + (-1) \times 3 + 0 \times 4 + 1 \times 5$$

$$= 8$$

$$G_y = (1 \times 3) + (2 \times 2) + (1 \times 5) + (0 \times 2) + (0 \times 3) \\ + (0 \times 4) + (-1 \times 3) + (-2) \times 4 + (-1 \times 5)$$

$$= -4$$

$$\text{gradient magnitude} = \sqrt{64 + 16} = \sqrt{80} = 8.9$$

For (3,3) example

$$g_x = (-1 \times 2) + (0 \times 5) + (1 \times 1) + (-2 \times 3) + (0 \times 4) \\ + (2 \times 5) + (-1 \times 4) + (1 \times 6)$$

$$= 5$$

$$g_y = -7$$

gradient magnitude

$$= \sqrt{25 + 49} \\ = \sqrt{74} \\ = 8.6$$

15) Is there any object in size of an image by color quantization. Justify.

Solution

Yes, color quantization affects the size of an image because it reduces the number of unique colors used in the image, which leads to a reduction in file size.

- i) Reduction in bits per pixel (bpp)
- ii) Compression Efficiency.
- iii) Effect on lossy compression.

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